

CARINSPECT 2.0: Usability Study

March 2023

Focus: Brake & Tire

UX Researcher: Martin Losoviz

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01 Study Details

Project Background:

- 1.1. **Business Problem:** Inspecting vehicles is the single most important job we perform when identifying vehicle defects. The process is labor intensive, monotonous, and littered with inaccuracies.
- 1.2. **Solution:** A user centric inspection app to help our users make the right decisions when they are inspecting a vehicle reducing reworks and improving accuracy.

STUDY DETAILS

Research

How are users able to perform B&T tasks with the proposed prototype A?

What can we learn from the user flow for B&T that users take?

Layout A vs. Layout B

Participants

B&T Inspectors, Line Leads and Data entry Associates with high School degree between the ages of 20-40

Additional Participants:

Stakeholders, GMs and Managers

Research

Length:

10-15 minutes per participant

Location: US - Phoenix, Tolleson, Trenton, Elyria

Methodology: Moderated and Unmoderated testing:

Users were asked to perform a series of tasks on a low fidelity prototype

PROTOTYPES TESTED

A

CarInspect 2001234567

SPECS 2 UNDERCARRIAGE 1 B&T 1/4

DF PF

Tires PADS ROTORS

- 7 +

Done

Torque (ft-lb): 129
Tire size: P255/35/21

PF

Tires PADS ROTORS

- 7 +

Done

DR PR

Tires PADS ROTORS

- 7 +

Done

Tire Depth (2/32" Minimum)

☐ Dually

DF - Tire Passed
Inspection Step: PF - Tire
Reason: Above Spec
Part Type: Tire

SUBMIT

B

TIRES

Tires Thread

☐ DF 9 9 ☐ PF

☐ DR 9 9 ☐ PR

OPTIONS

PADS

PADS

☐ DF 9 9 ☐ PF

☐ DR 9 9 ☐ PR

OPTIONS

ROTORS

ROTORS

☐ DF 9 9 ☐ PF

02 Themes

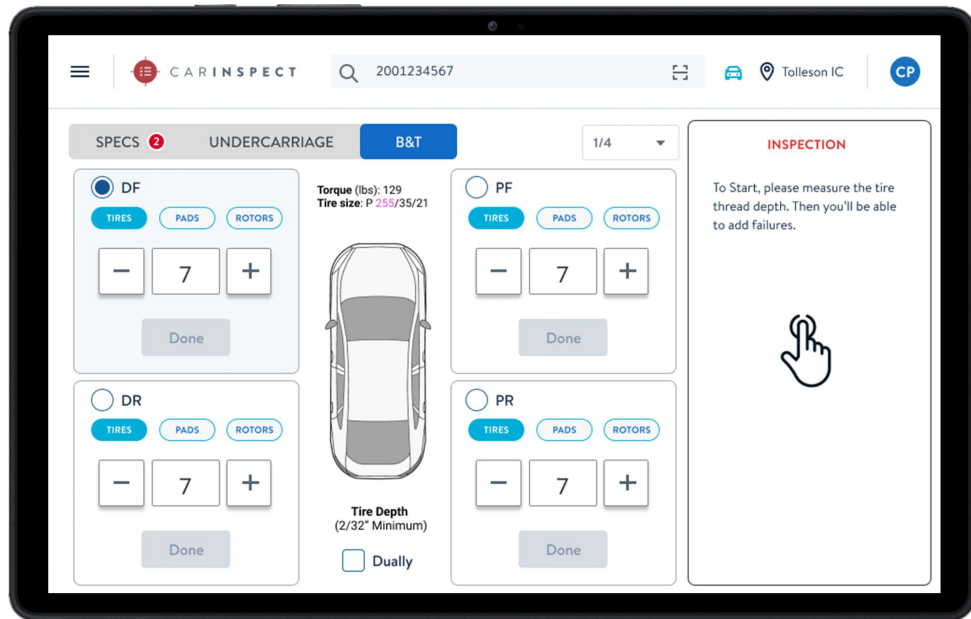
Most Participants want to be able to perform tire inspections in a quick and simple manner.

All users were able to move around the different corner tiles and add failures for different tires.

Most users think Layout A is simple, immersive and self explanatory.

“It’s pretty easy, it’s self explanatory for someone new who’s jumping in for the first time”

A

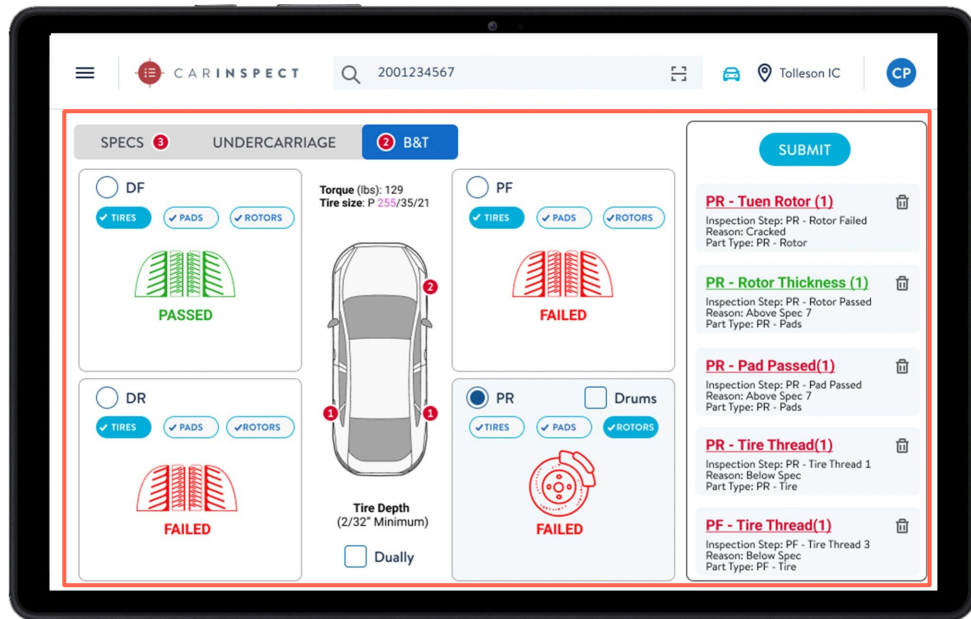


Most Participants want to perform all B&T actions in a single screen.

Most users prefer all actions to be performed on a single screen.

“It’s easy enough since everything happens in the same page, and you can see the status of what failed what hasn’t failed right there”

A

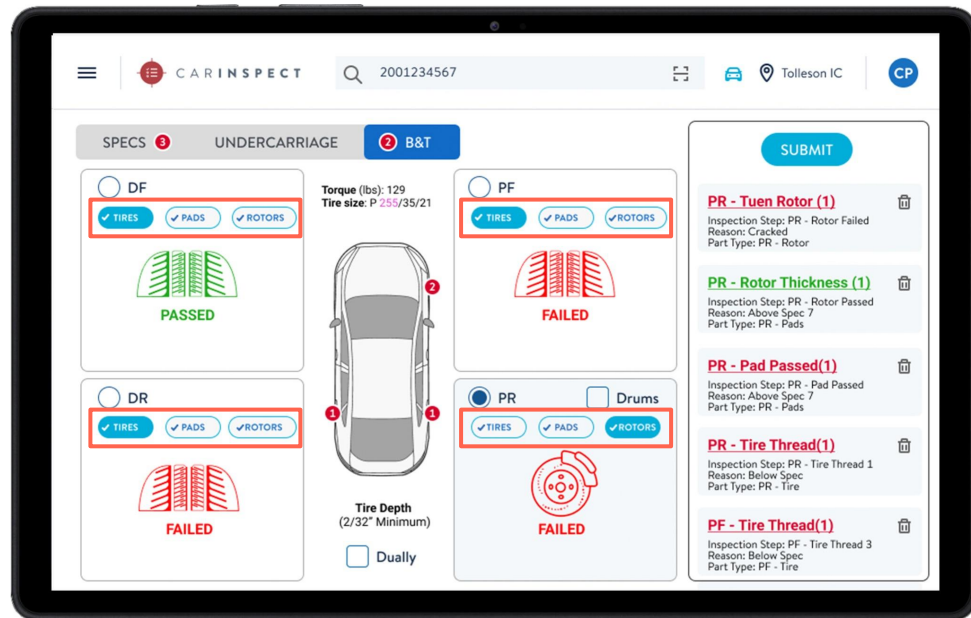


Users want to move quickly between tires pads and rotors.

A

Most users were able to fail tires, rotors and pads on all sides of the vehicle.

Some users expressed some frustration with the amount of clicks to perform a task.



Inspectors want to annotate a failure followed by a reason

Most users were able to add a failure followed by a specific reason.

Most users were able to associate a photo to a failure after a visual prompt was displayed.

A

FAILURE REASON

DR TIRE

☐ Below Spec

☐ Brand Mismatch

☐ Sidewall Damage

☐ Dry Rot

☐ Puncture

☐ Uneven Wear

☐ Wrong Direction

☐ Wrong Tire Size

Next

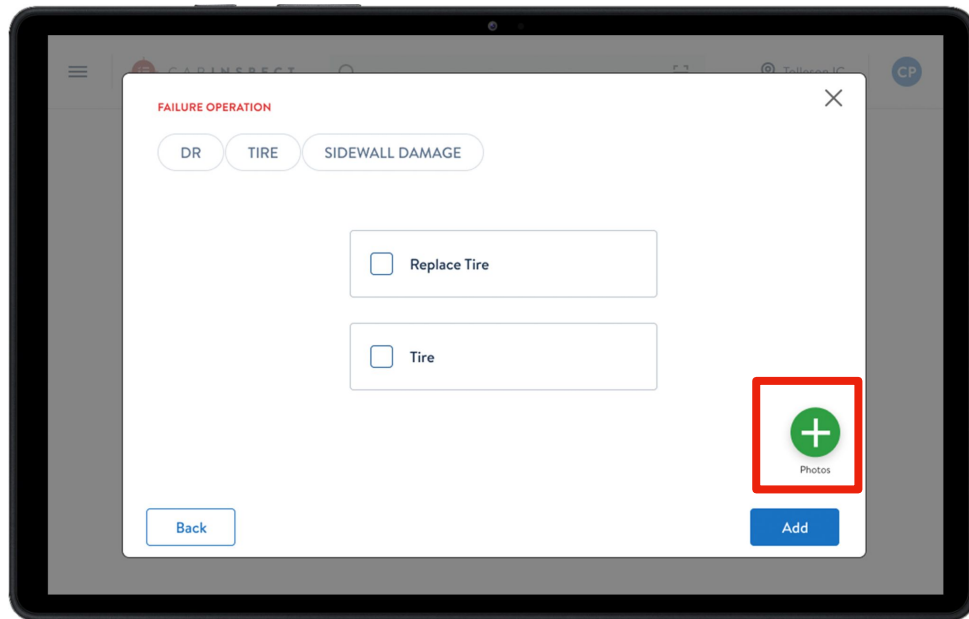
Users want to take photos while performing an inspection.

Most users were able to add a photo to a failure.

Some users were able to take a photo after a visual prompt was displayed.

Most users missed the “add photo” photo button.

A



Most users think that performing tire inspections is simple and self explanatory.

Most users expressed some level of frustration with Layout B.

Most users find that scrolling is less desirable.

“I feel some sections would get missed by the inspectors, if you have to scroll down”

B

TIRES

Thread

DF 9 9 PF

DR 9 9 PR

OPTIONS

PADS

PADS

DF 9 9 PF

DR 9 9 PR

OPTIONS

ROTORS

ROTORS

DF 9 9 PF

Users want to perform B&T actions in a single screen

Most users prefer all actions happening in a single screen.

Most users showed some level of frustration when scrolling to perform tasks.

“I feel this layout would eat a couple seconds of my time, while the other one A, I pretty much stay within the same screen and everything happens there”

B

The image shows a tablet displaying a mobile application interface. The interface is divided into three main sections, each with a header and a dropdown arrow: **TIRES**, **PADS**, and **ROTORS**. Each section contains a sub-header **Thread**. Below the sub-header, there are two rows of input fields. Each row consists of a checkbox, a text input field containing the number '9', another text input field containing the number '9', and a final checkbox. The checkboxes are labeled 'DF', 'PF', 'DR', and 'PR' respectively. Below each row of input fields is a red button labeled **OPTIONS**. The layout is consistent across all three sections.

03 Key Findings

KEY Findings

A vs. B Testing

Most users
Prefer layout A
vs B. They prefer
the visual and
immersive
layout rather
than the long
scroll.

Reduce amount of steps

Most users want
to be able to
add a fail in less
number of
clicks.

Consistency

Most users rather
have a consistent
way to input
values in the
system for
Tire/Pads/Rotors

Unable to attach photos

some users
missed the
photo button
but were able to
take photo after
being
prompted.

Multi- Select

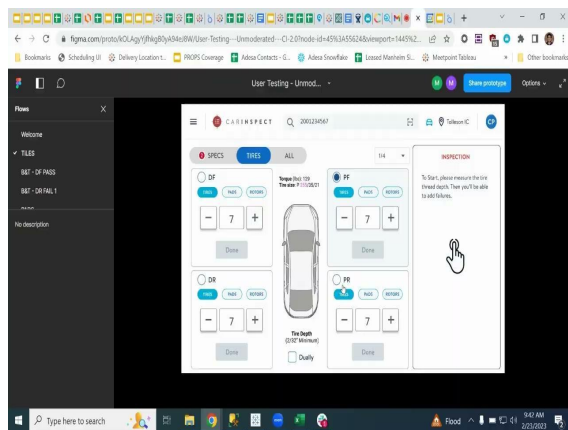
Some users
would like to fail
multiple parts at
the same time.

RECOMMENDATIONS

- 1.1. Iterate on B&T design A to be able to fail a part in less steps.
- 1.2. Make all input values consistent across all B&T and CarInspect
- 1.3. Make the photo action button more discoverable
- 1.4. Consider adding a multi-select option for bulk fail.

Videos

Michael



Alfred



Mickey



THANK YOU!